P1748R1
Fill in [delay.cpp] TODO in P1389

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This version:
https://yehezkelshb.github.io/cpp_proposals/sg20/P1748-fill-delay-cpp-section-in-P1389.html

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SG20

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ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Source:
GitHub

Abstract

P1748 adds wording to replace the TODO of [delay.cpp] section in P1389

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§ 1. Revision History

r0: initial revision, pre-Cologne mailing

r1:

- update "generic function" example based on feedback from sg20 in Cologne
- style update

§ 2. Proposed Wording

Under 2.3.5 C Preprocessor [**delay.cpp**]:

~~Excludes `#include`, which is necessary until modules are in C++.~~

~~TODO~~

Most of the traditional usages of the C Preprocessor have better and safer C++ replacements. For example:

```
// compile-time constant
#define BUFFER_SIZE 256
// better as:
auto constexpr buffer_size = 256;

// named constants
#define RED   0xFF0000
#define GREEN 0x00FF00
#define BLUE  0x0000FF
// better as:
enum class Color { red = 0xFF0000, green = 0x00FF00, blue = 0x0000FF };

// inline function
#define SUCCEEDED(res) (res == 0)
// better as:
inline constexpr bool succeeded(int const res) { return res == 0; }

// generic function
#define IS_NEGATIVE(x) ((x) < 0)
// better as:
template <typename T>
bool is_negative(T x) {
   return x < T{};
}
```

All these macros have many possible pitfalls (see gcc docs), they hard to get right and they don’t obey scope and type rules. The C++ replacements are easier to get right and fit better into the general picture.

The only preprocessor usages that are necessary right at the beginning are:

- `#include` for textual inclusion of header files (at least until modules become the main tool for consuming external code)
- `#ifndef` for creating "include guards" to prevent multiple inclusion

§ 3. Acknowledgements